

HW # 4

u 2210 251

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$$h(x, y, z) = \ln(3x + 6y + 9z) \quad (1, 1, 1)$$

$$v = 4i + 12j + 6k$$

$$h(x, y, z) = \left\langle \frac{3}{3x+6y+9z}, \frac{6}{3x+6y+9z}, \frac{9}{3x+6y+9z} \right\rangle$$

$$h(1, 1, 1) = \left\langle \frac{1}{6}, \frac{1}{3}, \frac{1}{2} \right\rangle$$

$$u = \frac{1}{\sqrt{16+144+36}} (4i + 12j + 6k) = \frac{2}{7}i + \frac{6}{7}j + \frac{3}{7}k$$

$$D_u h(1, 1, 1) = h(1, 1, 1) \cdot u = \frac{1}{21} + \frac{2}{7} + \frac{3}{14} = \frac{33}{42}$$

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$$f(p, q, r) = \arctan(pqr) \quad (1, 2, 1)$$

$$f(p, q, r) = \left\langle \frac{qr}{1+(pqr)^2}, \frac{pr}{1+(pqr)^2}, \frac{pq}{1+(pqr)^2} \right\rangle$$

$$f(1, 2, 1) = \left\langle \frac{2}{5}, \frac{1}{5}, \frac{2}{5} \right\rangle$$

$$\text{Max} \Rightarrow |f(1, 2, 1)| = \sqrt{\frac{4}{25} + \frac{1}{25} + \frac{4}{25}} = \frac{3}{5}$$

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$$f(x, y) = ye^{-xy}$$

$$f_x(x, y) = ye^{-xy}(-y) = -y^2 e^{-xy}$$

$$f_y(x, y) = ye^{-xy}(-x) + e^{-xy} = (\cancel{xy} - xy)e^{-xy}$$

$$f_x(0, 2) = -4 \quad f_y(0, 2) = 1$$

$$D_u f(0, 2) = f_x(0, 2) \cos \theta + f_y(0, 2) \sin \theta =$$

$$= -4 \cos \theta + \sin \theta$$

$$1 + \cos^2 \theta + 8 \cos \theta = 8$$

$$-4 \cos \theta + \sin \theta = 1$$

$$\cos \theta (1 + \cos \theta + 8) = 0$$

$$\sin \theta = 1 + 4 \cos \theta$$

$$\cos \theta = 8 \quad \theta = \frac{\pi}{2}$$

$$\sin^2 \theta = (1 + 4 \cos \theta)^2$$

$$\cos \theta = -\frac{8}{17} \quad \theta = \frac{3\pi}{2}$$

$$1 - \cos^2 \theta = 1 + 8 \cos \theta + 16 \cos^2 \theta$$

$$\theta = \frac{3\pi}{2}$$

no

$$\theta = \frac{\pi}{2}$$

$$\theta = 2\pi - \cos^{-1}\left(-\frac{8}{17}\right) \approx 4.22 \text{ rad}$$

[page 368 exercise 41]

$$2(x-2)^2 + (y-1)^2 + (z-3)^2 = 10 \quad (3, 3, 5)$$

$$F(x, y, z) = 2(x-2)^2 + (y-1)^2 + (z-3)^2$$

$$F_x(x, y, z) = 4(x-2) \quad F_y(x, y, z) = 2(y-1)$$

$$F_x(3, 3, 5) = 4 \quad F_y(3, 3, 5) = 4$$

$$F_z(x, y, z) = 2(z-3) \quad F_z(3, 3, 5) = 4$$

a) $(3, 3, 5)$ at $4(x-3) + 4(y-3) + 4(z-5) = 0$

by eq 18 $4x + 4y + 4z = 44$

$$x + y + z = 11$$

b) $\frac{x-3}{4} = \frac{y-3}{4} = \frac{z-5}{4}$

by eq 20

$$x-3 = y-3 = z-5$$

$$x = 3+t \quad y = 3+t \quad z = 5+t$$

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$$f(x, y) = x^2 + y^2 - 2x \quad \Rightarrow (2, 0) \text{ and } (0, -2)$$

$$f_x = 2x - 2 \quad f_y = 2y$$

$$f_x = f_y = 0 \Rightarrow (1, 0)$$

$$f(1, 0) = -1$$

$$L_1: x=0 \quad f(0,y)=y^2 \quad \text{for } -2 \leq y \leq 2$$

$$\min \Rightarrow y=0 \quad f(0,0)=0$$

$$\max \Rightarrow y=\pm 2 \quad f(0,\pm 2)=4$$

$$L_2: y=x-2 \quad 0 \leq x \leq 2 \quad \left| \begin{array}{l} f(x,x-2)=x^2-6x+4= \\ \min \Rightarrow x=\frac{3}{2} \quad f(\frac{3}{2},-\frac{1}{2})=-\frac{1}{2} \end{array} \right. \quad \left| \begin{array}{l} =2(x-\frac{3}{2})^2-\frac{1}{2} \\ \max \Rightarrow x=0 \quad f(0,-2)=4 \end{array} \right.$$

$$\min \Rightarrow x=\frac{3}{2} \quad f(\frac{3}{2},-\frac{1}{2})=-\frac{1}{2}$$

$$\max \Rightarrow x=0 \quad f(0,-2)=4$$

$$L_3: y=2-x \quad 0 \leq x \leq 2 \quad f(x,2-x)=x^2-6x+4=$$

$$\min \Rightarrow x=\frac{3}{2} \quad f(\frac{3}{2},\frac{1}{2})=-\frac{1}{2} \quad \left| \begin{array}{l} =2(x-\frac{3}{2})^2-\frac{1}{2} \\ \max \Rightarrow x=0 \quad f(0,2)=4 \end{array} \right.$$

$$\max \Rightarrow x=0 \quad f(0,2)=4$$

$$\text{abs. max of } f \text{ on } D \Rightarrow f(0,\pm 2)=4$$

$$\text{abs. min of } f \text{ on } D \Rightarrow f(1,0)=-1$$

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$$2^2 = x^2 + y^2 \quad (4, 2, 0)$$

$$d = \sqrt{(x-4)^2 + (y-2)^2 + 2^2}$$

$$d^2 = (x-4)^2 + (y-2)^2 + x^2 + y^2 = f(x,y)$$

$$f_x(x,y) = 2(x-4) + 2x = 4x-8 \quad \text{critical point } (2,1)$$

$$f_y(x,y) = 2(y-2) + 2y = 4y-4 \quad \text{closest } (2,1,\pm\sqrt{2})$$

$$f_x \Rightarrow 0 \Rightarrow x=2 \quad f_y \Rightarrow 0 \Rightarrow y=1$$